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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
December 2024

ME1654-1 – INNOVATION AND DESIGN THINKING
(For AD, BT, CB, CS, CV, EE, RI)

Duration: 2 Hours

Max. Marks: 50

Note: Answer Three full questions choosing One full question from each Unit

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Define design process and also write the requirements for a good design solution.	7	L*1	1	1
	b) Explain, briefly the various phases of design thinking approach.	8	L2	1	1
2.	a) Explain creativity, invention and innovation processes using a common example.	6	L1	1	1
	b) How do you empathetically collect data from a person who is looking for a hat/cap for picnic, sports and trekking applications using 360 degree design thinking approach.	9	L2	1	1
Unit – II					
3.	a) Compare agile model and waterfall model with sketches and explain.	8	L2	2	1
	b) Write the advantages of agile model.	7	L1	2	1
4.	a) Define business process model and also explain its advantages.	7	L1	2	1
	b) Explain in brief the steps involved in scenario based prototyping.	8	L2	2	1
Unit – III					
5.	How do you empathize, define and ideate a backpack (bag) for a person who wants to use it for his daily use, hiking application that includes a night stay and bike riding using 360degree design thinking approach. Assume suitable inputs wherever necessary.	20	L3	3	2
6.	How do you empathize, define and ideate a water bottle for a person who wants to use it for his daily use, gymnastic purpose and trekking applications using 360degree design thinking approach. Assume suitable inputs wherever necessary.	20	L3	3	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
